

SMART INDIA HACKATHON 2026

POLARIS: Integrated Polar Expedition Operations & Asset Suite



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Problem Statement ID – SIH26062

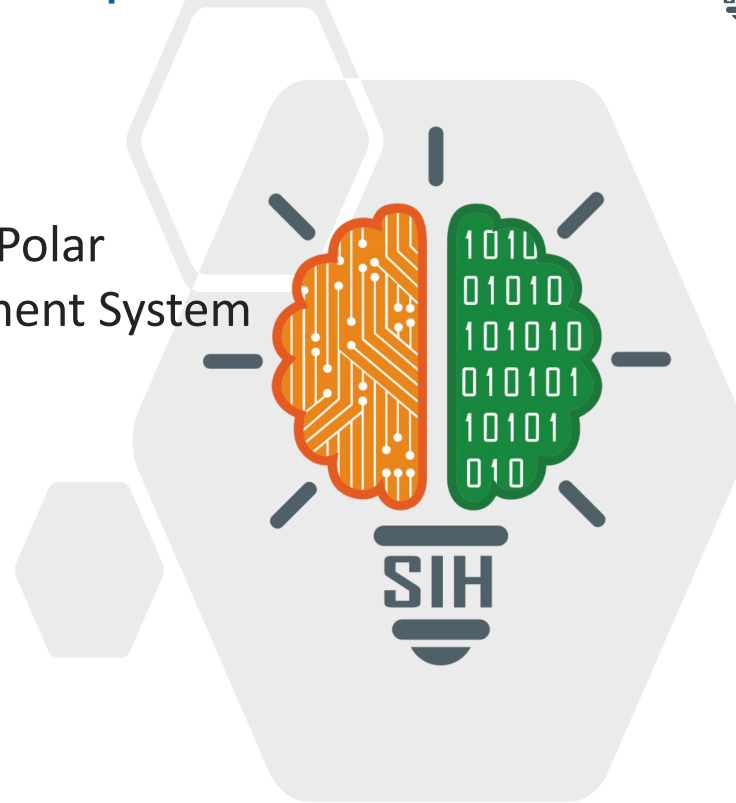
Problem Statement Title – Integrated Polar Expedition Logistics and Asset Management System

Theme – Smart Automation

PS Category – Software

Team ID – 137578

Team Name (Registered on portal) – Dev React





POLARIS: Centralized Polar Expedition ERP & Asset Operations Suite



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Proposed Solution Overview

Centralized Operations Hub:

Connects **NCPOR Goa** with Antarctic stations (**Bharati**, **Maitri**), Arctic base (**Himadri**), and icebreaker **MV Vasily Golovnin**.

4 Core Integration Pillars:

1. Expedition Voyage Scheduling
2. Cold-Chain Asset Inventory
3. Personnel Safety & Roster
4. Blizzard Emergency Response

Operational Impact:

- Accelerates ice-edge unloading reconciliation from **72h to <6h**.
- Prevents life-support failures during 14-month polar night overwintering cycles.

Core Operational Modules

VOYAGE PLANNER

Coordinates chartered vessel routes (**Goa → Cape Town → Ice Shelf**), cargo payload balancing, and mission Gantt milestones.

COLD-CHAIN ERP

Tracks Arctic diesel, Jet A-1 fuel, freeze-dried rations, and snowcat (PistenBully) spares with sub-zero shelf-life monitoring.

PERSONNEL SAFETY & ROSTER

Digital station muster rolls, field camp geofencing (Schirmacher Oasis, Larsemann Hills), and daily health clearances.

EMERGENCY SOS & HAZARD COCKPIT

Automated 3-tier blizzard threat tracking (Condition 1/2/3) & 1-click COSPAS-SARSAT distress beacon dispatch.

★ Key Uniqueness & Impact

100% Offline-First Resilience

- Operates completely offline during polar satellite dropouts.
- Uses local-first architecture to persist state seamlessly.
- Auto-merges conflict-free when satellite links flicker back on.

Predictive Burn-Rate Modeling

- Correlates ambient sub-zero temperatures (-40°C) with active headcount.
- Predicts critical resource & fuel exhaustion months in advance.
- Ensures uninterrupted station life support.



TECHNICAL APPROACH & SYSTEM ARCHITECTURE



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STAGE 1: DATA INGESTION

Edge Sensing & Capture

- IoT RFID & Cargo Tags
- Station Telemetry Feeds
- Satellite Drop-link Telemetry
- Local Edge Field Capture (<5ms IndexedDB persist)



STAGE 2: POLARIS ENGINE

Local-First Core (Offline Sync)

- IndexedDB 100% Offline Cache
- CRDT Conflict-Free Replicated State
- Store-and-Forward Sync Queue
- FastAPI & SQLite Station Edge



STAGE 3: COMMAND & CONTROL

Operational Analytics

- Predictive Fuel & Burn-Rate ML
- Vessel Voyage Dispatch & Route
- COSPAS-SARSAT SOS Beacon
- Hazard Indexing & Lockout Rules

CORE TECH STACK & FRAMEWORKS

React 18

PWA UI Framework

Vite

Ultra-fast Build Engine

Python (FastAPI)

Backend & Scikit-Learn ML

Tailwind CSS

Responsive Styling

Web Crypto API

Encrypted Edge Sync



FEASIBILITY ANALYSIS & POLAR RISK MITIGATION



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1. EXTREME SUB-ZERO ENVIRONMENT (-40°C) ENVIRONMENTAL RISK

Challenge: Severe sub-zero conditions risk drive freezing, thermal contraction, and hardware failures on station edge units.

→ Industrial-Grade Offline RFID & Ruggedized UI

Low-overhead containerized standards (Docker, SQLite, PWA) operating reliably on rugged tablets without specialized procurement.

2. POLAR SATELLITE BLACKOUTS CONNECTIVITY RISK

Challenge: Extended ionospheric blizzards and solar storms frequently sever polar satellite links for weeks.

→ Conflict-Free Offline PWA with Background Sync

Store-and-forward queue lets local nodes operate autonomously indefinitely; safely syncing in micro-bursts upon link restoration.

3. LIFE-SUPPORT FUEL / RATION DEPLETION CRITICAL SAFETY RISK

Challenge: Unplanned inventory consumption during 14-month polar night overwintering cycles.

→ Predictive ML Burn-Rate Early Warning

Dual-node active-passive edge clustering at Bharati & Maitri with automated hourly encrypted flash backups and burn forecasts.

4. MULTI-STATION INTEROPERABILITY GOVERNANCE & COMPLIANCE

Challenge: Paper manifests, disconnected Excel sheets, and legal liabilities for polar environmental governance.

→ Unified ERP Spanning Antarctica & Arctic

Air-gapped sovereign design with Madrid Protocol environmental ledger for automated tracking of hazardous waste & fuel bunkering.



STRATEGIC, ECONOMIC & LIFE-SAFETY IMPACT



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PILLAR 1: STRATEGIC AUTONOMY

ATMANIRBHAR

POLAR GOVERNANCE FOR MoES / NCPOR

Sovereign Operational Backbone

Delivers an indigenously developed polar operational framework, elevating India's standing at ATCM meetings.

360° Operational Visibility

Continuous situational awareness over station occupancy, fuel health, and scientific field traverses.

Madrid Protocol Compliance

Automated tracking of hazardous waste, fuel bunkering, and mandatory return-waste back to India.

PILLAR 2: OPERATIONAL EFFICIENCY

72h → <6h

12X FASTER ICE-EDGE RECONCILIATION

Rapid Icebreaker Offload

Slashes chartered vessel cargo manifest reconciliation time on fast ice shelf from 72h to <6h.

Supply Chain Loss Prevention

Prevents multi-crore losses from misplaced machinery spares and damaged perishable rations.

Air-Charter Optimization

Eliminates costly emergency air-charters from Cape Town via accurate burn-rate forecasting.

PILLAR 3: LIFE SAFETY & ZERO FAILURE

< 2 MIN

INSTANT TRIAGE & TARGET ZERO CASUALTIES

Rapid Search & Rescue (SAR)

Slashes blizzard SAR dispatch latency from hours to under 2 minutes with automated distress beacons.

Geofenced Field Tracking

Real-time tracking and automated weather lockouts prevent field disorientation during whiteouts.

14-Month Life Support Reserve

Protects wintering crews by ensuring unbroken power generator fuel reserves through polar nights.



RESEARCH FOUNDATIONS, STANDARDS & REFERENCES



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OFFICIAL INSTITUTIONAL PORTALS

[ncpor.res.in] → National Centre for Polar and Ocean Research

MoES expedition logistics & station records

[comnap.aq] → Council of Managers of National Antarctic Programs

Station safety & air operations

[ats.aq] → Antarctic Treaty Secretariat

Madrid Environmental Protocol Annex III & IV

[scar.org] → Scientific Committee on Antarctic Research

Polar environmental & data standards

TECHNICAL & PROTOCOL REFERENCES

[cospas-sarsat.int] → International Satellite SAR

406 MHz satellite Search & Rescue distress beacon specifications

[epsg.io/3031] → EPSG:3031 Polar Coordinate Reference

WGS 84 Antarctic Polar Stereographic geospatial projection

[mqtt.org / [ISO:20922](https://www.iso.org/standard/62448.html)] → Lightweight Messaging

Lightweight messaging transport standard for low-bandwidth satellite links

[localfirstweb.com] → ACM Research: Local-First Software

Local-First Software & Conflict-Free Replicated Data Types (CRDTs)